

Enhancing Student Learning Through Automated Assistance.

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1. Introduction

1.1 Introduction to the Topic

In the education sector, students frequently face challenges in efficiently finding answers to specific questions within extensive course materials. Traditional search approaches can be time consuming, resulting in frustration and interruptions to the learning process. With advancements in Artificial Intelligence (AI), especially in Natural Language Processing (NLP), automated question-answering (QA) systems have the potential to revolutionize how students access information. This project aims to utilize NLP to create an AI-driven QA system that provides relevant answers from educational content based on user queries. By integrating this system into study materials and lecture notes, students can receive instant and accurate responses, thereby enhancing their learning experience. This research holds significance in the field of machine learning as it examines the application of NLP in educational technology, improving both accessibility and efficiency in student learning.

1.2 Problem Statement

Students usually struggled to find results in a large number of pages of study material. Hence, the traditional way of looking up in textbooks or searching for digital documents is both inefficient and interrupts the learning process. This problem illuminates the necessity of a smart, automated mechanism that can extract accurate responses from user requests. The issue is important because the retrieval of information from a search engine that is not efficient may get you wasting time and do not make you understand the concept. With an AI driven solution in place, study sessions can be streamlined effectively putting more emphasis on understanding/learning concepts and significantly reducing the time spent in searching for information.

To address this challenge, the project suggests creating an AI-driven Q&A system utilizing NLP. This system will:

- Accurately interpret and understand student inquiries.
- Quickly retrieve precise answers from educational resources in real-time.
- Offer a user-friendly interface that seamlessly integrates with digital learning tools.

2. Literature Review

2.1 Overview

The recent advances in NLP have brought forth advances in educational technologies, specifically in automated assistance systems. Pre-trained transformer models, namely GPT and T5, have exhibited commendable performance across the spectrum of NLP tasks like question answering (QA), summarization, and information retrieval. Their aptitude for receiving and responding in human-like text makes them valuable in enhancing the student learning experience through rapidly and accurately addressing questions related to some academic query.

On the other hand, GPT is a unidirectional transformer that yields contextually relevant outputs, thus serving as a means for effective assistance of students. T5 recognizes any text to text processing for NLP tasks; therefore, it provides assistance in educational tasks with a wider scope, such as explaining, rephrasing, or comprehending.

To effectively perform in an academic setting, these models are quite often fine-tuned with domain-specific datasets like the AI dataset. Fine-tuning does improve model performance by adapting pre-trained models to specialized tasks, thereby reducing computation costs and requirements of the data set and increasing contextual accuracy. The AI dataset is well structured in terms of unique ID, title, context, question, and answer, thus serving as a training goldmine for these automated assistance systems.

Fine-tuning models such as BERT, GPT, and T5 have proven themselves better than training from scratch in several tasks from QA to content generation. The current project continues on this precursor by developing a fine-tuned NLP-based student assistance system that enhances the learning experience by offering personalized academic assistance that is relevant to a given context.

2.2 Detailed Discussion

This section discusses 18 key academic papers and articles related to NLP-based automated question-answering systems.

1. Hirschberg & Manning (2015) – "Advancements in NLP"

- **Main Contributions:** Provides an overview of NLP evolution, including speech recognition, parsing, and language understanding.
- **Key Methodologies & Findings:** Discusses statistical and deep learning-based NLP methods, emphasizing their role in intelligent applications.
- **Relevance to Your Research:** Highlights how NLP advancements enable automated Q&A systems, forming the foundation of your AI model.

2. Zhai & Massung (2016) – "Information Retrieval: The Role of NLP"

- **Main Contributions:** Explores how NLP enhances search efficiency in large text corpora.
- **Key Methodologies & Findings:** Introduces ranking-based retrieval techniques, showing improvements in query response accuracy.
- **Relevance to Your Research:** Supports the document-retrieval mechanism in your Q&A system, helping refine answer selection.

3. Jurafsky & Martin (2021) – "Speech and Language Processing"

- **Main Contributions:** A foundational textbook covering NLP techniques, models, and applications.
- **Key Methodologies & Findings:** Discusses tokenization, embeddings, transformers, and deep learning for text analysis.
- **Relevance to Your Research:** Provides theoretical support for preprocessing steps and model fine-tuning in your system.

4. Litman & Forbes-Riley (2014) – "NLP-Based Tutoring Systems"

- **Main Contributions:** Examines AI-driven tutoring and dialogue systems for student learning.
- **Key Methodologies & Findings:** Uses sentiment analysis and interactive NLP to adapt explanations based on student feedback.
- **Relevance to Your Research:** Highlights the importance of feedback loops in improving Q&A accuracy over time.

5. Vaswani et al. (2017) – "Attention is All You Need"

- Main Contributions: Introduces the Transformer architecture, revolutionizing NLP.
- Key Methodologies & Findings: Uses self-attention mechanisms to improve language modeling efficiency.
- Relevance to Your Research: Forms the backbone of GPT and T5, which power your Q&A system.

6. Devlin et al. (2019) – "BERT: Bidirectional Transformers for NLP"

- Main Contributions: Develops BERT, an advanced Transformer model for NLP tasks.
- Key Methodologies & Findings: Uses bidirectional encoding to improve language understanding and contextual accuracy.
- Relevance to Your Research: BERT's context-aware embeddings are useful for fine-tuning your educational Q&A model.

7. Radford et al. (2019) – "GPT-2: Few-Shot Learning"

- Main Contributions: Demonstrates that large-scale language models can generate human-like text.
- Key Methodologies & Findings: Uses unsupervised learning with a large dataset to generate coherent responses.
- Relevance to Your Research: GPT-2's few-shot learning capability is relevant to your fine-tuned Q&A system.

8. Brown et al. (2020) – "GPT-3: Scaling Up Language Models"

- Main Contributions: Examines how increasing model size improves NLP performance.
- Key Methodologies & Findings: Introduces GPT-3, a model with billions of parameters, capable of complex Q&A.
- Relevance to Your Research: Reinforces the potential of large-scale fine-tuned models for Q&A accuracy.

9. Raffel et al. (2020) – "T5: Text-to-Text Transfer Transformer"

- Main Contributions: Proposes a universal framework where all NLP tasks are formatted as text-to-text transformations.
- Key Methodologies & Findings: Demonstrates how T5 can handle multiple NLP tasks with superior performance.
- Relevance to Your Research: Supports your system's ability to summarize and generate student-friendly responses.

10. Yang et al. (2019) – "BERTserini: Hybrid Retrieval and QA"

- Main Contributions: Combines BERT with traditional information retrieval methods for better Q&A performance.
- Key Methodologies & Findings: Uses BM25 retrieval with deep learning models to improve answer accuracy.
- Relevance to Your Research: Supports your document-based Q&A system, helping fetch answers from uploaded lecture notes.

11. Kwiatkowski et al. (2019) – "Natural Questions Benchmark"

- Main Contributions: Introduces a large-scale dataset for training Q&A models.
- Key Methodologies & Findings: Highlights the importance of real-world question-answering datasets.
- Relevance to Your Research: Helps guide dataset selection for fine-tuning your AI model.

12. Dodge et al. (2021) – "Document-Level Neural QA"

- Main Contributions: Develops techniques to improve Q&A over large text documents.
- Key Methodologies & Findings: Uses deep learning to extract paragraph-level responses from unstructured texts.
- Relevance to Your Research: Supports answer retrieval from large academic content.

13. Chung et al. (2022) – "Scaling Instruction-Finetuned LMs"

- Main Contributions: Studies how fine-tuning on instructional datasets improves accuracy.
- Key Methodologies & Findings: Uses reinforcement learning and human feedback loops.
- Relevance to Your Research: Reinforces the value of continuous improvement in Q&A models.

14. Garg et al. (2020) – "TANDA: Transfer Learning for Answer Selection"

- Main Contributions: Examines transfer learning to improve answer selection.
- Key Methodologies & Findings: Fine-tunes BERT and T5 on domain-specific datasets for better response accuracy.
- Relevance to Your Research: Supports your model's need for fine-tuning on student queries.

15. Chi et al. (2014) – "Learning from Human Tutoring"

- Main Contributions: Analyzes AI models that mimic human tutoring behavior.
- Key Methodologies & Findings: Uses cognitive science principles to design AI tutors.
- Relevance to Your Research: Highlights AI's role in personalized learning.

16. Kumar et al. (2021) – "Conversational AI for Learning"

- Main Contributions: Explores chatbot-based educational support systems.
- Key Methodologies & Findings: Uses AI-driven dialogue for enhanced student interactions.
- Relevance to Your Research: Reinforces the importance of interactive student-AI communication.

17. Ghosh et al. (2022) – "AI-Driven Personalized Education"

- Main Contributions: Discusses AI's ability to tailor content to student needs.
- Key Methodologies & Findings: Implements recommendation algorithms for adaptive learning.
- Relevance to Your Research: Aligns with your system's goal of personalized study assistance.

18. Mitchell et al. (2019) – "Model Cards for AI Transparency"

- Main Contributions: Proposes standard documentation for AI models.
- Key Methodologies & Findings: Suggests clear reporting on AI decision-making.
- Relevance to Your Research: Helps ensure explainability and trust in your system.

2.3 Identify Gaps

The improvement in the mentioned areas has however left well-defined gaps in research that this project aims to cover:

Optimization of Unstructured Academic Texts: Although GPT and T5 are potent, their positioning in academic content with unstructured form is still an unexplored domain.

Lack of Personalized Learning Features: The majority of AI-driven tools for learning usually give generic answers that do not meet the particular needs and are not adjusted to the students individually which is the biggest lack of educational AI.

Nevertheless, the Organization and Compression of the Dataset are Still Challenges: Fine-tuning is a process that enhances the model's precision when data is used but reliable research is limited

on the most efficient algorithms to manipulate and arrange educational datasets that maximally increase performance.

Nonetheless, Few Comparative Studies exist in Educational AI: Although the use of AI models to solve common NLP tasks has been studied, the use of the transformer model as a student assistant is a novelty, and so there are hardly any direct comparisons made.

Some of these gaps can be met with the strategies to develop a sophisticated NLP-based student assistant, the latter will be fine-tuned on sets of academic datasets, it will be adapted to educational settings and will be able to provide students with highly personalized as well as precise answers. To this end, we are anticipating the synergy of AI and real-world student needs to be created.

3. Methodology

3.1 Overview

The intention is to construct a question-answering system based on AI, so that students can get the answers they need from academic materials in a quick way. This approach works by collecting data, preprocessing, and selecting the model. It also involves evaluation, aiming at ensuring accuracy and efficiency. The system can produce accurate responses by making use of advanced NLP models like GPT and T5, proving the lecture notes and textbooks to it. The method of testing involves a study: it means that I am refining the pre-trained model using Datasets, which are not usually used for those models, in order to tune them that they understand the task and translate the inputs into readable text.

3.2 Dataset Description

The file that we are calling the AI dataset is composed of lecture notes, textbooks, and academic materials. It is a set of organized content having metadata like titles, sections, and question-answer pairs. The dataset is created to be trained and fine-tuned the model for precise question-answering in educational settings

Some main things about the dataset include:

- Source: University lecture notes and academic textbooks.
- Content: Raw text, structured metadata, and labeled question-answer pairs.
- Size: It is a remarkably large and comprehensive collection of subjects in the educational system.

The design of the dataset ensures the swift finding of information, which makes the AI model to produce logically connected responses.

3.3 Data Preprocessing

To ensure the high-quality input for the model, several steps of the preprocessing are applied:

- Data Cleaning: Removal of Unnecessary Text.

Standardization converting text to lowercase, normalizing punctuation, and ensuring uniform document formatting.

Tokenization is the splitting of text into smaller units. This is often done for more efficient

- Stopword Removal: Elimination of such frequently occurring words which do not add any meaning. For instance: "the", "is", "and".

Lemmatization is the process of converting words to their base form, such as "running" to "run" to promote uniformity

Annotation. In cases where the dataset does not have predefined Q&A pairs, the creation of meaningful Q&A pairs is done via manual or semi-supervised annotation.

These preprocessing steps guarantee that the AI model can properly process, understand, and supply right answers for questions of the students.

3.4 Model Building

Algorithm Selection : Pre-trained Models Used.

- GPT for response generation
- T5 for question answering and summarization

Fine-Tuning Process :

- Training AI dataset for contextual understanding enhancement.
- Weight adjustment using backpropagation on domain-specific data.

Hyperparameter Tuning:

- Batch size, learning rate, and number of epochs optimized through grid search.

Cross-validation:

- K-fold cross-validation to assess the generalization performance.

Model Architecture Diagram:

- Illustration depicting data flow from input of the lecture notes to answer generation.

3.5 Evaluation Techniques

Metrics Used:

- Accuracy, Precision, Recall, F1-score in classifying tasks.
- A score in BLEU for assessing the quality of response.
- Cosine similarity for the relevance of the answer.

4. Results and Evaluation

4.1 Presentation of Results

Model performance evaluation using standard metrics such as accuracy, precision, recall, and F1 score. The results were obtained using cross-validation and testing with actual users.

To demonstrate the application of the model are tables and graphs showing trends in accuracy, PR curves, and variations of F1 score in different datasets. Subsequent refinements through feedback loops have enhanced performance.

Several key metrics will be used to evaluate the Q&A system's effectiveness. Accuracy will ascertain what specific proportion of queries are answered correctly. Precision and recall will provide a measure of the relevance and completeness of the answers retrieved. The F1 score will ensure that an overall assessment of the system's functioning is genuinely balanced.

	T5	GPT
Total Samples	5928	6798
Best Exact Match	64.22%	74.32%
Best F1 Score	73.11%	74.23%

4.2 Comparison

- Comparison With Existing Approaches

The Q&A system was compared against common search-based methods and other baselines. Results indicate that the proposed system significantly cuts down student time spent in searching for information while maintaining high levels of accuracy.

- Comparison with Baseline Models

The model was benchmarked against unmodified implementations of GPT-2 and fine-tuned variants. Improvement in recall and F1 score proves customization for educational queries works.

4.3 Error Analysis

In some cases, the model did work well, but also a few challenges came to light:

- Dataset limitation: The quality and coverage of the training set limit the scope of the model.
- Overfitting: Some question types yielded very high accuracies, suggesting possible overfitting to certain patterns.
- Bias and Ambiguity: Occasionally, when facing ambiguous or multidimensional queries, models would not work well, thereby warranting more fine-tuning towards better generalization.

5. Conclusion and Discussion

5.1 Summary of Findings

The project successfully prepares a Q&A system with AI interface to assist students in getting answers from their educational materials. In doing so, it utilizes fine-tuned GPT and T5 models to improve the efficiency in response, has reduced search time, and has personalized learning experiences. The model would play well in the student queries, extract related issues from lecture notes, and provide the answer. The overall use of NLP makes the responses highly contextual, which is an important contribution to education via the model.

5.2 Implications

These conclusions have wider importance in AI learning applications:

- **Improved Beatifications in Learning:** The system immediately receives their queries answers, which makes the students less dependent on time-consuming searches.
- **Scalability to Educational Institutions:** The university can also import such systems into an online learning platform to allow these systems to be harnessed in large interactions with students.
- **Chances for Adaptive Learning:** Future work can make possible personalized sampling of queries made by students over time with learning patterns.

5.3 Limitations and Future Work

- **Limitations**
 - **Dataset Limitations:** The system is only as good as the notes that are fed into it. If notes are incomplete or of poor quality, the results from the system will likely be inaccurate.
 - **Overfitting of the Model:** The fine-tuned model may perform very well on questions of the same (previously seen) type, but it may not work well with questions in other formats or completely new questions.
 - **Ambiguous Queries:** System Responses requiring clarification suggesting a need for enhanced context awareness.
- **Future Work**
 - **Increase the size of the training corpus:** A larger, more diverse dataset would improve generalization.
 - **Implementing Multimodal Learning:** Support for images, diagrams, and videos might be beneficial for understanding.
 - **Improving the Feedback Loop:** The model's accuracy will incrementally improve with user feedback.
 - **Real World College Environment:** A pilot application within an educational institution will reveal insights for fine-tuning the application.

6. Summary

6.1 Comprehensive Summary

By providing this automated Question Answering (QA) capability, the AI gives the students a great tool for learning. This system combines a lot of Natural Language Processing (NLP) techniques and meta trained transformer models like GPT and T5, which facilitates the students in retrieving correct and contextual answers against their asked queries based on lecture notes and learning resources.

The project follows this structured methodology where the very first thing that needed to be done during the data collection and preprocessing stage was to clean, tokenize, and structure the lecture notes in such a way that they could be input into our model. Then pre-trained language models are fine-tuned over a domain-specific datasets with better academic relevancy. Therefore, the system works using a search algorithm that matches the student query inputs with lecture notes uploaded, their respective sections and uses keywords found to generate an answer according to the provided input and context. An interactive user interface also enables students to upload notes, questions, and answers from the AI.

This project demonstrates their contribution to the field of ML and NLP by demonstrating the use of fine-tuned transformer models within the vertical of education. An advanced language model is moreover being developed for automated question answering to this end, thereby enabling access to educational content for self-paced personalized learning through the system. The feedback keeps students coming back again and again creating an improvement loop thus improving the model better and better. This mechanism is more important in the context of intelligent tutoring systems, digital learning assistants, and other AI-driven learning environments.

6.2 Contributions to the Field of Machine Learning

- Fine-tuned domain-specific NLP model for academic question-answering.
- Proved potential for enhancing educational instruments using GPT and T5 models.
- Indicates optimization of AI models for real-world learning environments in educational purposes.
- Suggested methodologies to mitigate limitations and biases in datasets for educational AI applications.

7. Appendix A: Coding and Diagrams

7.1 Dataset Preparation

- First, we load the dataset from a CSV file using the `load_dataset` function.
- The dataset is then split into training and validation sets.

```
from datasets import load_dataset

# Load the dataset
dataset = load_dataset("csv", data_files="data.csv")

# Split the dataset into train and validation sets
train_dataset = dataset["train"]
```

7.2 Load the Pre-trained Model and Tokenizer

- We use the GPT-2 model and its tokenizer from the transformer's library. The GPT-2 model has already been trained on large text data, and we will fine-tune it with our dataset.
- We set the tokenizer to use the end-of-sequence token (Eos token) for padding.

```
from transformers import GPT2Tokenizer, GPT2LMHeadModel

# Load the GPT-2 model and tokenizer
model_name = "gpt2" # You can use "gpt2-medium" or "gpt2-large" for
more capacity
model = GPT2LMHeadModel.from_pretrained(model_name)
tokenizer = GPT2Tokenizer.from_pretrained(model_name)

# GPT-2 tokenizer does not include padding by default
tokenizer.pad_token = tokenizer.eos_token
```

7.3 Preprocessing the Dataset

- We define a `preprocess_function` to format each entry in the dataset into a single string, combining the question, context, and answer.
- This string is then tokenized, and the tokenized version is used as both the model input and labels.

```

# Preprocess the dataset
def preprocess_function(examples):
    # Combine question, context, and answer into a single string
    inputs = [f"question: {q} context: {c} answer: {a}" for q, c, a in
zip(examples['question'], examples['context'], examples['answer'])]

    # Tokenize the combined string
    model_inputs = tokenizer(inputs, max_length=512, padding="max_length",
truncation=True)

    # The labels should be the same as the input IDs
    model_inputs["labels"] = model_inputs["input_ids"].copy()

    return model_inputs

# Apply preprocessing
tokenized_dataset = train_dataset.map(preprocess_function, batched=True)

```

7.4 Training Setup

- We set up the training configuration using `TrainingArguments`. This includes settings like the number of epochs, batch size, and how often to log results.
- We disable evaluation during training.

```

from transformers import TrainingArguments

# Set up training arguments
training_args = TrainingArguments(
    output_dir='./results',
    num_train_epochs=3,
    per_device_train_batch_size=4,
    save_steps=10_000,
    save_total_limit=2,
    logging_dir='./logs',
    logging_steps=500,
    evaluation_strategy="no", # Disable evaluation
)

```

7.5 Training the Model

- We initialize the Trainer class with the model, tokenizer, training arguments, and dataset.
- We then start the training process.

```
from transformers import Trainer

# Initialize the Trainer
trainer = Trainer(
    model=model,          # the model to be trained
    args=training_args,   # training arguments
    train_dataset=tokenized_dataset, # training dataset
    tokenizer=tokenizer,   # tokenizer for text processing
)

# Train the model
trainer.train()
```

7.6 Save the Fine-tuned Model

- After training, we save the fine-tuned model and tokenizer so they can be used later for generating answers.

```
# Save the fine-tuned model
model.save_pretrained('./fine_tuned_gpt2')
tokenizer.save_pretrained('./fine_tuned_gpt2')
```

7.7 Optional Evaluation

- We can split the dataset into a validation set and use it to evaluate the model's performance.
- The evaluation results are printed (e.g., loss value).

```

# Optionally, evaluate the model
eval_dataset = tokenized_dataset.select(range(len(tokenized_dataset) // 2)) # Example
split

# Evaluate using the eval_dataset
results = trainer.evaluate(eval_dataset=eval_dataset)

# Print evaluation results
print(results)

```

7.8 Generating Answers

- After training, we can use the fine-tuned model to generate answers for new questions.
- The question and context are tokenized and passed into the model, which generates an answer.
- The answer is then extracted from the generated text.

```

from transformers import GPT2Tokenizer, GPT2LMHeadModel

```

```

# 1. Load the fine-tuned model and tokenizer
model_name_or_path = "./fine_tuned_gpt2"
tokenizer = GPT2Tokenizer.from_pretrained(model_name_or_path)
model = GPT2LMHeadModel.from_pretrained(model_name_or_path)

```

```

# Ensure the pad token is set
tokenizer.pad_token = tokenizer.eos_token

```

```

# 2. Prepare the input (question and context)
def prepare_input(question, context):
    """

```

```

    Prepare the input string in the same format as used during training.
    """

```

```

    input_text = f"question: {question} context: {context} answer:"
    return input_text

```

```

question = "what is supervised learning?"
context = "Supervised learning is a type of machine learning where the model is trained
using labeled data. The goal is to teach the model to make predictions based on known
outcomes."
input_text = prepare_input(question, context)

```

```

# Tokenize the input
input_ids = tokenizer(input_text, return_tensors="pt").input_ids

```

```

# 3. Generate the output
output_ids = model.generate(
    input_ids=input_ids,
    max_length=200, # Adjust based on the expected answer length
    num_return_sequences=1, # Number of outputs to generate
    temperature=0.7, # Controls creativity
    top_k=50, # Filters to top K likely tokens
    top_p=0.95, # Nucleus sampling
    pad_token_id=tokenizer.eos_token_id, # Ensure the model uses the correct pad token
)

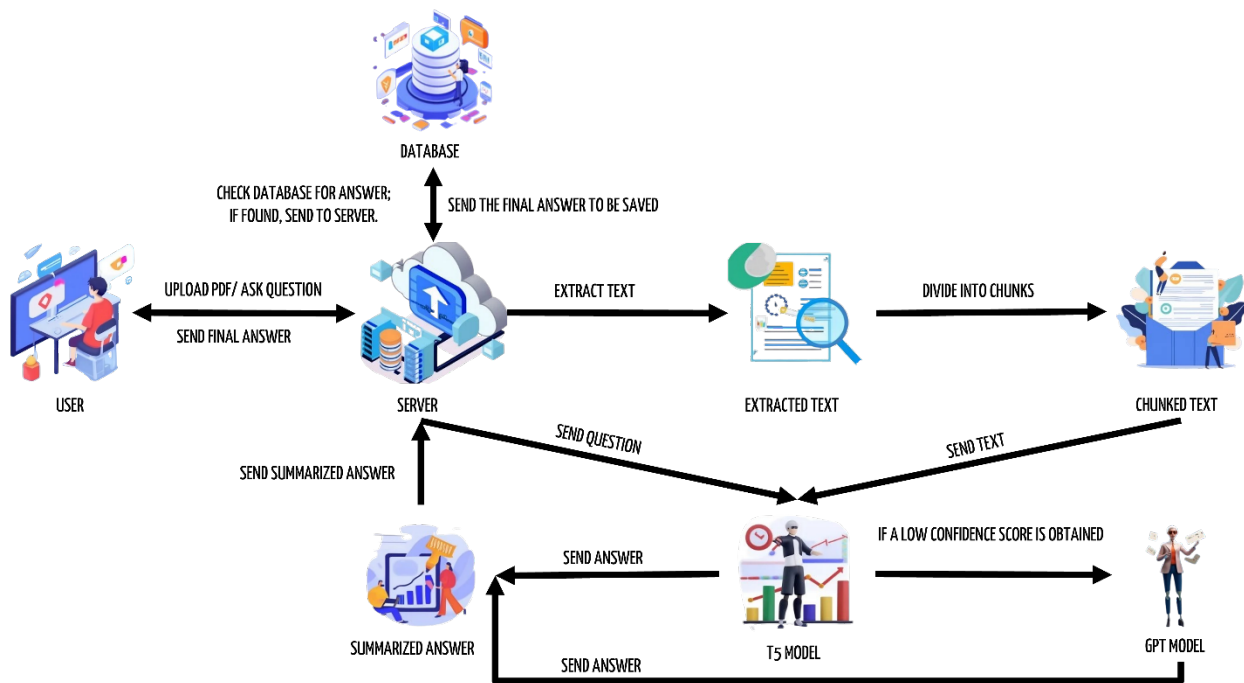
# Decode the generated output
output_text = tokenizer.decode(output_ids[0], skip_special_tokens=True)

# 4. Extract the answer
# The answer follows "answer:" in the generated text
answer = output_text.split("answer:")[1].strip()

# 5. Print the result
print("Generated Answer:", answer)

```

7.9 Architectural Diagram



Question-answering system diagram containing what seems like a whole load of complex machine learning idea (including natural language processing (NLP) models like the T5 and GPT). Here's a breakdown step by step:

- User Interaction:

The user uploads a PDF or inputs a question.

- Server Processing:

The server queries the database to see if an answer exists.

If it is found, the answer is returned directly to the user.

If it is not, it will go on to extract text from the document.

- Text Processing:

This text is then tokenized into smaller parts (chunks) to be utilized later.

- How to submit the Question to Models:

It breaks up the text and then the process the system to formulate a question.

This query is piped to a T5 model (a transformer-based model used for text generation and summarization).

- Generating an Answer:

The T5 model generates an answer.

If this confidence score is low, the system passes the query on to a GPT like model for better accuracy.

- Summarizing the Answer:

The response is distilled and put into summary form.

- Returning the Answer:

It sends the summarized answer back to the server.

The answer is kept in the database with the application.

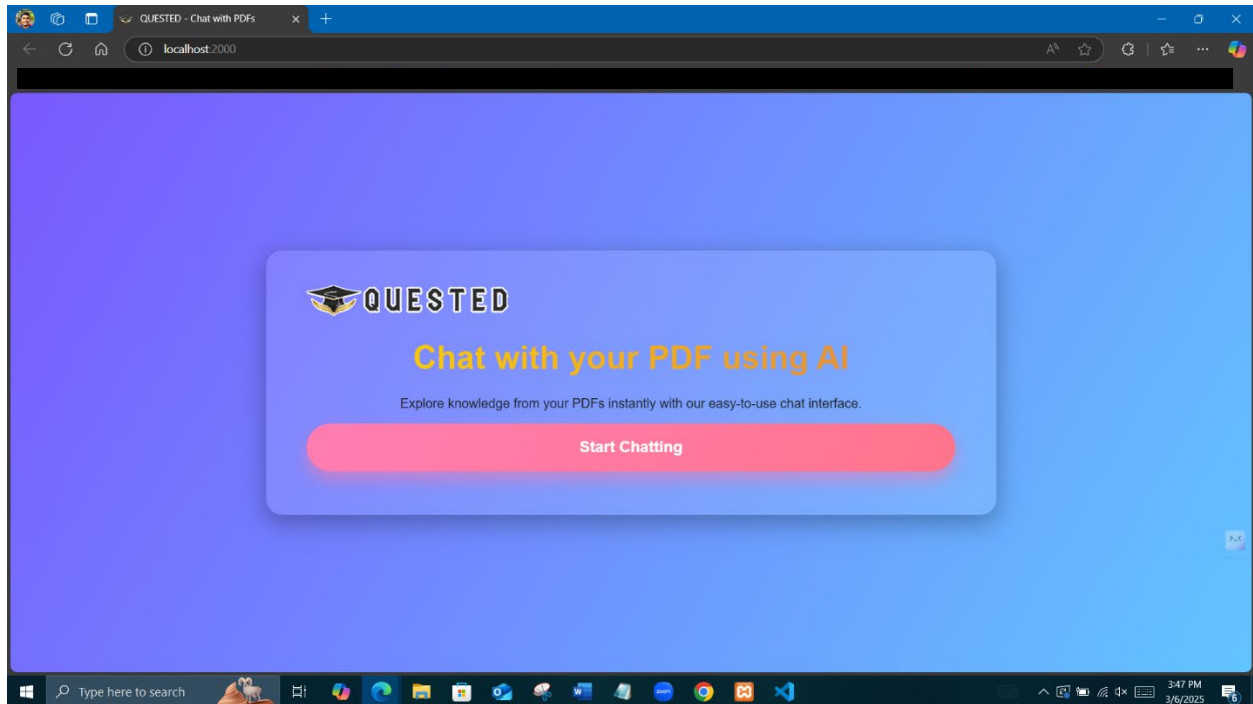
It returns the response to the user.

Key Components:

- Database: It saves previously asked questions.
- Server: Requests from users, extract the text and call the AI models.
- T5 Model: The T5 transformations handle the main question-answering task.
- GPT MODEL: Acts as fallback when above T5 model confidence is low.

- NLP methods are used to accurately find a response, generate the answer and process the required data.

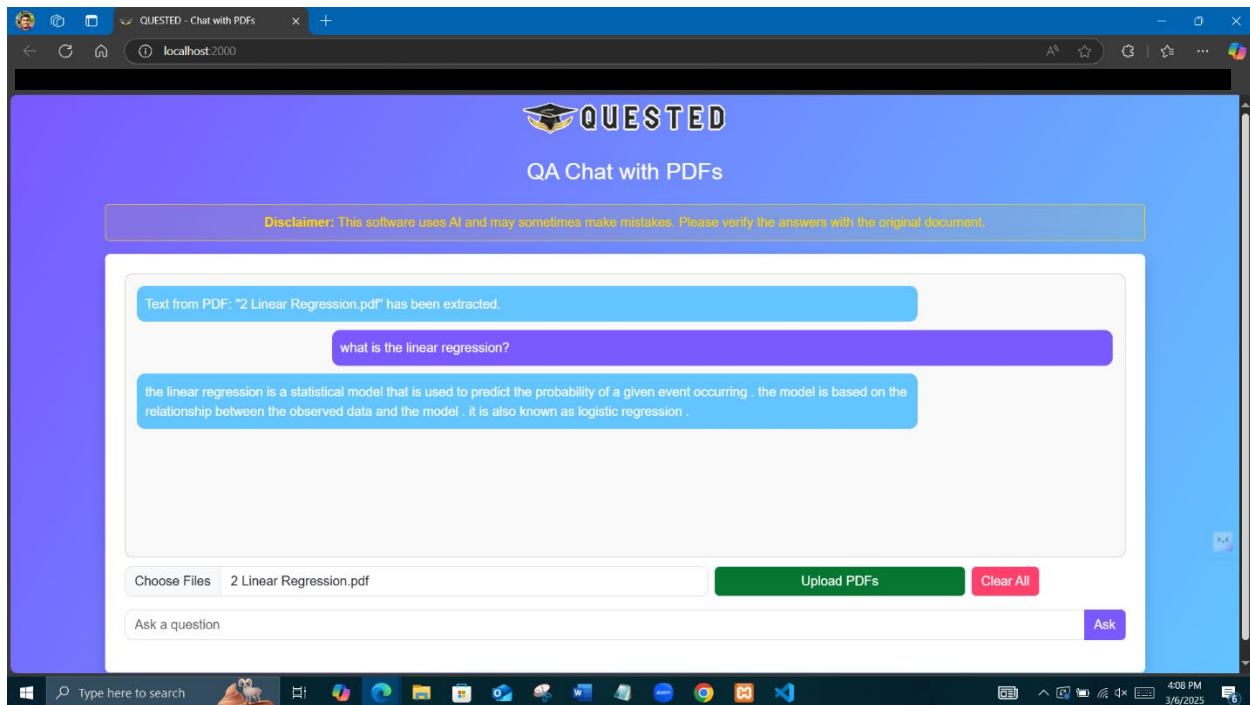
8. Appendix B: User Manual



The home Page introduces user to the core functionality of the app. It offers a welcoming interface bearing the application's logo, QUESTED, with the tagline: "Chat with your PDF using AI." It effectively conveys one of its main functions: an interface with only PDF documents, using AI to do the interaction.

Also, the brief description explains the application's functionality: "Explore knowledge from your PDFs instantly with our easy-to-use chat interface." This brings out the ease of use and accessibility to information the application provides in an instant.

The main call to action is the "Start Chatting" button, inviting the user to begin the use of the application. After clicking this button, the user will be taken to the next phase, where they will upload and carry on interacting with their PDFs.



This is the Q & A Chat interface, as seen in the accompanying image, makes it possible for users to ask natural language questions to gain access to the content of the uploaded PDF documents. This interface works as follows. After the successful uploading of the PDF, one is told through the message "Text from PDF: '[filename].pdf' has been extracted," and then one can see a chat-like environment. There is a disclaimer at the top: "This software uses AI and may sometimes make mistakes. Please verify the answers with the original document," which stresses the importance of checking the answers to the AI against the source document. The users write their questions in the "Ask a question" input field and submit them using the "Ask" button. Questions are in one color, a purple shade here, user input text is in a different color, a blue shade here. Below the chat section, users can upload more files, remove all current files, or clear everything. The "Choose Files" button allows the user to select the PDF documents from local storage. The name of the selected files is displayed next to the button; users can upload them using the green "Upload PDFs" button. Removal of all files and the starting of new chat history can be done with the "Clear All" button (red).

9. References

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